

Prioritized Embedded To-Do List with Algorithms and Data Structures

This embedded to-do list is prioritized by difficulty, incorporating real-time data sources, algorithms, and data structures.

1 Priority 1: Foundation-Level Tasks

1.1 1. Temperature and Voltage Monitoring

- **Data Source:** Internal temperature sensor, voltage reference (VREFINT)
- **Data Structures:**
 - Ring Buffer (store recent sensor data)
 - Priority Queue (track highest/lowest readings)
- **Algorithms:**
 - Moving Average/Exponential Smoothing
 - Threshold Detection Algorithm

1.2 2. Watchdog Timers

- **Data Source:** Independent Watchdog Timer (IWDG), Window Watchdog Timer (WWDG)
- **Data Structures:**
 - Circular Queue (log reset events)
- **Algorithms:**
 - Timeout Management Algorithm
 - Monitoring and Alerting Algorithm

1.3 3. Real-Time Clock (RTC) and Timestamps

- **Data Source:** RTC, SysTick timer
- **Data Structures:**
 - Binary Search Tree (store and query timestamps)
 - B-tree (for large timestamp sets)
- **Algorithms:**
 - Scheduling Algorithms (round-robin, priority-based)
 - Binary Search (query nearest event)

2 Priority 2: Intermediate-Level Tasks

2.1 4. Interrupt-Driven Task Scheduling

- **Data Source:** Timers, interrupts
- **Data Structures:**
 - Priority Queue (schedule tasks by priority)
 - Doubly Linked List (dynamic scheduling)
- **Algorithms:**
 - Round-Robin Scheduling
 - Earliest Deadline First (EDF)

2.2 5. Real-Time Signal Processing with ADC

- **Data Source:** Analog-to-Digital Converter (ADC)
- **Data Structures:**
 - Circular Queue (buffer real-time data)
 - FFT Buffer (for frequency-domain analysis)
- **Algorithms:**
 - Fast Fourier Transform (FFT)
 - Sliding Window Algorithm (compute statistics over time)
 - Finite State Machine (FSM) for pattern recognition

2.3 6. Fault Monitoring and Recovery

- **Data Source:** Non-Maskable Interrupt (NMI), system faults
- **Data Structures:**
 - HashMap (categorize and track faults)
 - Circular Queue (log recent fault events)
- **Algorithms:**
 - Error Detection and Correction (Hamming Code)
 - Cyclic Redundancy Check (CRC)

3 Priority 3: Advanced-Level Tasks

3.1 7. DMA for Data Streaming

- **Data Source:** DMA controller, ADC
- **Data Structures:**
 - Ring Buffer (buffer incoming data)
 - Trie (manage packet sequences or binary streams)
- **Algorithms:**
 - Pattern Matching Algorithms (KMP, Rabin-Karp)
 - Circular Buffer Management (real-time data handling)

3.2 8. Memory Management and Flash Programming

- **Data Source:** Flash memory, EEPROM
- **Data Structures:**
 - HashMap (store key-value pairs in flash)
 - Linked List (store history of write operations)
- **Algorithms:**
 - Wear-Leveling Algorithm (distribute flash writes)
 - CRC for Data Integrity Verification

3.3 9. Power Consumption Monitoring

- **Data Source:** Internal power sensors
- **Data Structures:**
 - Priority Queue (track power events)
 - HashMap (map tasks to power usage)
- **Algorithms:**
 - Power Optimization Algorithms (dynamic voltage/frequency scaling)
 - Energy Profiling Algorithms (track usage over time)